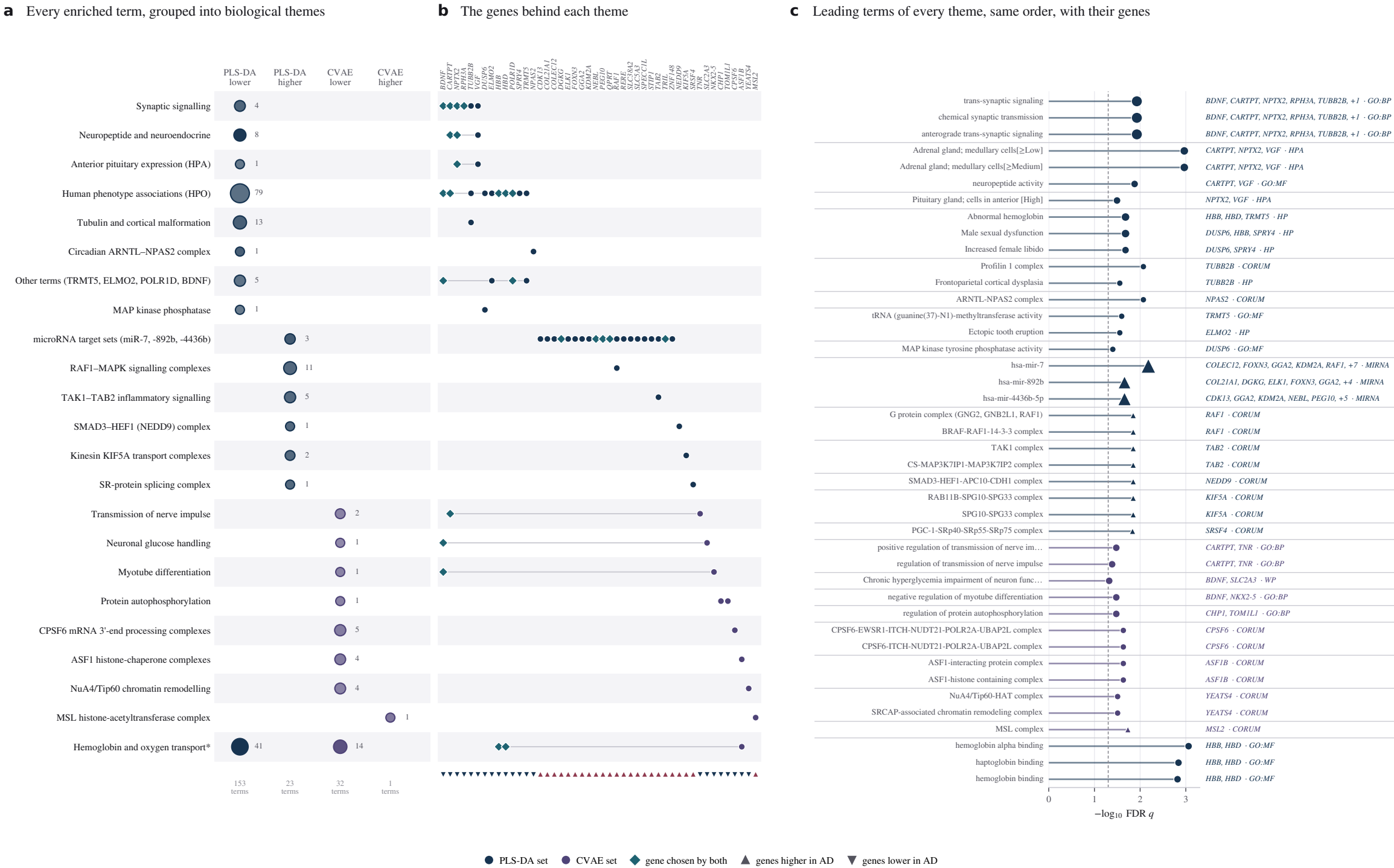
a Every enriched term, grouped into biological themes
b The genes behind each theme
c Leading terms of every theme, same order, with their genes
PLS-DA lower
PLS-DA higher
CVAE lower
CVAE higher
Synaptic signalling
Neuropeptide and neuroendocrine
Anterior pituitary expression (HPA)
Human phenotype associations (HPO)
Tubulin and cortical malformation
Circadian ARNTL–NPAS2 complex
Other terms (TRMT5, ELMO2, POLR1D, BDNF)
MAP kinase phosphatase
microRNA target sets (miR-7, -892b, -4436b)
RAF1–MAPK signalling complexes
TAK1–TAB2 inflammatory signalling
SMAD3–HEF1 (NEDD9) complex
Kinesin KIF5A transport complexes
SR-protein splicing complex
Transmission of nerve impulse
Neuronal glucose handling
Myotube differentiation
Protein autophosphorylation
CPSF6 mRNA 3'-end processing complexes
ASF1 histone-chaperone complexes
NuA4/Tip60 chromatin remodelling
MSL histone-acetyltransferase complex
Hemoglobin and oxygen transport*
153 terms
23 terms
32 terms
1 terms
BDNF CARTPT NPTX2 RPH3A TUBB2B VGF DUSP6 ELMO2 HBB HBD POLR1D SPRY4 TRMT5 NPAS2 COL21A1 COLEC12 DGKG ELK1 FOXN3 GGA2 KDM2A NEBL PEG10 RAF1 RERE SLC38A2 SLC5A3 SPECC1L TTID TAB2 TRIL ZNF148 NEDD9 KIF5A SRSF4 TNR SLC2A3 NKX2-5 CHP1 TOM1L1 CPSF6 ASF1B YEATS4 MSL2
trans-synaptic signaling
chemical synaptic transmission
anterograde trans-synaptic signaling
Adrenal gland; medullary cells[≥Low]
Adrenal gland; medullary cells[≥Medium]
neuropeptide activity
Pituitary gland; cells in anterior [High]
Abnormal hemoglobin
Male sexual dysfunction
Increased female libido
Profilin 1 complex
Frontoparietal cortical dysplasia
ARNTL-NPAS2 complex
tRNA (guanine(37)-N1)-methyltransferase activity
Ectopic tooth eruption
MAP kinase tyrosine phosphatase activity
hsa-mir-7
hsa-mir-892b
hsa-mir-4436b-5p
G protein complex (GNG2, GNB2L1, RAF1)
BRAF-RAF1-14-3-3 complex
TAK1 complex
CS-MAP3K7IP1-MAP3K7IP2 complex
SMAD3-HEF1-APC10-CDH1 complex
RAB11B-SPG10-SPG33 complex
SPG10-SPG33 complex
PGC-1-SRp40-SRp55-SRp75 complex
positive regulation of transmission of nerve im…
regulation of transmission of nerve impulse
Chronic hyperglycemia impairment of neuron func…
negative regulation of myotube differentiation
regulation of protein autophosphorylation
CPSF6-EWSR1-ITCH-NUDT21-POLR2A-UBAP2L complex
CPSF6-ITCH-NUDT21-POLR2A-UBAP2L complex
ASF1-interacting protein complex
ASF1-histone containing complex
NuA4/Tip60-HAT complex
SRCAP-associated chromatin remodeling complex
MSL complex
hemoglobin alpha binding
haptoglobin binding
hemoglobin binding
BDNF, CARTPT, NPTX2, RPH3A, TUBB2B, +1 · GO:BP
BDNF, CARTPT, NPTX2, RPH3A, TUBB2B, +1 · GO:BP
BDNF, CARTPT, NPTX2, RPH3A, TUBB2B, +1 · GO:BP
CARTPT, NPTX2, VGF · HPA
CARTPT, NPTX2, VGF · HPA
CARTPT, VGF · GO:MF
NPTX2, VGF · HPA
HBB, HBD, TRMT5 · HP
DUSP6, HBB, SPRY4 · HP
DUSP6, SPRY4 · HP
TUBB2B · CORUM
TUBB2B · HP
NPAS2 · CORUM
TRMT5 · GO:MF
ELMO2 · HP
DUSP6 · GO:MF
COLEC12, FOXN3, GGA2, KDM2A, RAF1, +7 · MIRNA
COL21A1, DGKG, ELK1, FOXN3, GGA2, +4 · MIRNA
CDK13, GGA2, KDM2A, NEBL, PEG10, +5 · MIRNA
RAF1 · CORUM
RAF1 · CORUM
TAB2 · CORUM
TAB2 · CORUM
NEDD9 · CORUM
KIF5A · CORUM
KIF5A · CORUM
SRSF4 · CORUM
CARTPT, TNR · GO:BP
CARTPT, TNR · GO:BP
BDNF, SLC2A3 · WP
BDNF, NKX2-5 · GO:BP
CHP1, TOM1L1 · GO:BP
CPSF6 · CORUM
CPSF6 · CORUM
ASF1B · CORUM
ASF1B · CORUM
YEATS4 · CORUM
YEATS4 · CORUM
MSL2 · CORUM
HBB, HBD · GO:MF
HBB, HBD · GO:MF
HBB, HBD · GO:MF
0
1
2
3
−log10 FDR q
PLS-DA set
CVAE set
gene chosen by both
genes higher in AD
genes lower in AD
Bubble area = number of terms in the theme (count beside it); fill depth = best FDR q. Themes of one-gene terms are named by us from the terms' content. *Hemoglobin terms likely reflect residual blood and are not interpreted.